

1. Preprocessing Pipeline

The preprocessing pipeline cleans, validates, normalises, and structures incoming satellite imagery and text queries before any specialist vision-language model (VLM) forward pass occurs 1-3.

- **Data Ingestion & Quality-Scoring Filter (Training Phase):**
- **reBEN Ingestion:** Ingests the **BigEarthNet.txt** training dataset via reBEN (rico-hdl conversion) into an LMDB/safetensors store for high-throughput random-batch reads, using a geographic split to eliminate spatial train/test leakage 4-6.
- **ScoreRS Quality Filter:** Before fine-tuning, a learned quality-scoring model ranks training image-text pairs and discards the bottom quartile of noisy or low-quality annotations, improving downstream model performance 6-8.
- **Metadata Extraction & Compatibility Validation (Inference Phase):**
- **GDAL Format Validation:** Sniffs image headers to enforce allowed file formats (GeoTIFF, TIFF, PNG, JPEG) 9, 10.
- **Modality & Metadata Inspection:** Extracts band counts, Coordinate Reference Systems (CRS), nodata/cloud mask percentages (rejecting/flagging if nodata > 40%), Ground Sample Distance (GSD), and acquisition timestamps 3, 10, 11.
- **Geospatial & Temporal Precondition Checking:** Computes spatial footprint overlap via shapely Intersection over Union (IoU $\geq 70\%$), verifies date ordering for bi-temporal tasks, and checks CRS reprojectability via pyproj 10, 12.
- **Domain-Gap Mitigation & Sensor Normalization:**
- To bridge the 5–30x resolution gap between Sentinel-1/2 training data (10–60 m) and high-resolution evaluation sensors (**Cartosat-2S** optical at 0.65–2 m and **RISAT SAR**), the system applies 13-15:
- **GSD-Normalization Tiling:** Rescales spatial dimensions and tile sampling to standardize resolution scales 2, 14, 16.
- **Radiometric-Jitter Augmentation:** Performs histogram matching tailored to Cartosat-2S dynamic response curves and injects calibrated speckle noise matching RISAT SAR backscatter statistics 14, 16, 17.
- **Content-Addressed Embedding Caching:** Hashes uploaded raw raster bytes to cache visual-encoder embeddings, avoiding re-computation on repeat queries 18, 19.

2. Post-Processing Pipeline

After specialist models generate raw outputs, post-processing ensures outputs are geometrically valid, cross-verified, correctly calibrated for confidence, and serialized for auditability 1, 2, 20, 21.

- **Verifier Node (Geometric & Quantity Sanity Checks):**
- **Geometric Sanity:** Verifies bounding box coordinates lie strictly within normalized 22° image bounds, verifies oriented box angles fall within $(-90^\circ, 90^\circ)$, and ensures change-mask extents do not exceed co-registered footprint overlap boundaries 21, 23.
- **Cross-Tool Agreement:** Compares independent evidence sources (e.g., region tags from fusion vs. single-image VQA outputs) for semantic consistency 21.

- **Deterministic Quantity Cross-Check:** For counting/ratio questions, extracts deterministic instance/pixel counts from change maps and compares them against the LLM's verbalized numerical claims 21, 24, 25. Discrepancies exceeding a 15% tolerance band set a quantityDiscrepancy flag 21.
- **Confidence Scoring & Calibration:**
 - Uses a priority-ordered rule list to emit a **High**, **Medium**, or **Low** confidence badge (rather than raw probabilities) 26-28:
 - **Hard Override (Low):** Triggered if any geometric sanity check fails or a quantity discrepancy is detected 28.
 - **Structural Weak-Point Discount:** Applies a one-tier discount (e.g., High $\rightarrow$ Medium) for query types known to be empirical weak points (counting, ratio, smallest-change, non-unique referent grounding) 8, 26, 28.
- **Decoupled Perception/Reasoning Tokens:** Elicits separate internal confidence tokens for visual perception ("did I see correctly") vs. reasoning ("did I answer correctly") 26, 29, 30.
- **Response Composition & Visual Evidence Rendering:**
- **Natural Language Verbalization:** Synthesizes verified evidence ledgers into human-readable answers, explicitly citing which modality or specialist tool provided each claim 1, 16, 31.
- **Visual Map Overlays:** Renders bounding boxes (with scores and multi-candidate ambiguity styling) and change masks over Leaflet/deck.gl web viewer layers or GeoTIFF rasters 2, 32, 33.
- **Trace Serialization & Report Generation:**
- **ExecutionTrace Emitter:** Serializes step-by-step records containing selected tasks, invoked tools, adapter IDs/versions, runtime parameters, step timing (ms), and confidence rationales 1, 20, 22, 34.
- **PDF/JSON Report Service:** Compiles the trace, evidence images, and answers into a downloadable PDF/JSON report via weasyprint 2, 21, 35.

3. Optimizers & Training Hyperparameters

All specialized tasks share a single, compact **1–4B parameter open-weight**

Vision-Language Model (VLM) backbone (e.g., **InternVL3-1B-class** or small **Qwen-VL variant**) 8. The heavy vision encoder (e.g., CLIP-ViT-L/14) and base LLM decoder remain completely frozen 8, 36, 37.

Hyperparameter / Component, Value / Configuration

Fine-Tuning Method, "Low-Rank Adaptation (LoRA) 8, 37"

Trainable Parameters, "< 1%" of total model parameters 8, 36, 37"

LoRA Rank (r), "Rank 8 to 16 8, 37"

LoRA Target Modules, "Attention projection layers ((Q, K, V, O)) in the LLM decoder 8, 37"

Modality Projection Layers, "Linear projection layers for Sentinel-1 SAR and Sentinel-2 multispectral tokens 8, 37"

Optimizer, "AdamW (standard for LoRA VLM fine-tuning) 36, 37"

Task Conditioning Tokens, "Explicit task tokens prepended to instructions: [vqa], [caption], [ground], [change], [fusion] 38, 39"

Compute Budget, ~ 1 GPU-days per adapter on a modern multi-GPU node 39

4. Loss Functions

Training and aligning SatQuery AI's specialist adapters and fusion components utilizes four primary loss functions:

- **Autoregressive Token Cross-Entropy Loss ($\mathcal{L}_{\text{CE}}$):**
 - **Used for:** Single-image VQA, scene captioning, and change-VQA text generation 37, 40.
 - **Formulation:** Standard negative log-likelihood over target answer/caption token sequences: $\mathcal{L}_{\text{CE}} = -\sum_{t=1}^T \log P(w_t \mid w_{<t}, \mathbf{X}_{\text{vis}}, \mathbf{X}_{\text{text}})$
- **Sequence-to-Sequence Box Regression Loss ($\mathcal{L}_{\text{box}}$):**
 - **Used for:** Text-guided region grounding in adapter_grounding 37.
 - **Formulation:** Coordinates are tokenized into discrete text strings ($[x_{\text{Left}}, y_{\text{Top}}, x_{\text{Right}}, y_{\text{Bottom}}, \theta]$), casting spatial bounding box regression as autoregressive token cross-entropy over coordinate tokens 37.
- **Difference-Attention End-to-End Change Loss ($\mathcal{L}_{\text{change}}$):**
 - **Used for:** Bi-temporal change analysis in adapter_change_vqa 24, 40.
 - **Formulation:** Optimizes the Change-Enhancing Module (CEM)—which computes difference-attention over Siamese-encoded (T_1) and (T_2) feature representations—jointly with the LLM text decoder using end-to-end token cross-entropy 24, 40.
- **InfoNCE Contrastive Alignment Loss ($\mathcal{L}_{\text{InfoNCE}}$):**
 - **Used for:** Cross-modal SAR-to-RGB feature alignment within the Optical-SAR Fusion Pipeline's Complementarity Detector 15, 16, 29, 41.
 - **Formulation:** Aligns dense SAR-DINO and RGB-DINO feature representations across co-registered imagery (MM-OVSeg recipe) 16, 29, 41: $\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_{\text{sar}}, \mathbf{z}_{\text{rgb}}))}{\sum \exp(\text{sim}(\mathbf{z}_{\text{sar}}, \mathbf{z}_{\text{rgb}})) + \tau}$ (In ablation studies, InfoNCE contrastive alignment achieved **73.1% mIoU**, significantly outperforming MSE 67.7% and (L_1) loss 69.0% 15).

5. Evaluation Metrics & Benchmarks

SatQuery AI is evaluated across public open-source benchmarks and ISRO/SAC test sets 5, 42-44.

Benchmark Tasks & Evaluation Metrics

Task / Component	Target Dataset / Benchmark	Primary Evaluation Metric(s)
Single-Image VQA	RSVQA (LR/HR), VRSBench, ISRO/SAC Set 5, 42-44"	"VQA Accuracy (Overall, Binary, Closed-Set, Open-Ended) 37, 44"
Scene Description / Captioning	"VRSBench, BigEarthNet.txt, ISRO/SAC Set 5, 42-44", "BLEU-1/4, CIDEr, METEOR, ROUGE-L 37, 44"	
Text-Guided Region Grounding	"VRSBench, ISRO/SAC Set 5, 42-44", " $(\text{Acc})@0.5$ (IoU (≥ 0.5)), Mean Average Precision (mAP) 8, 37, 45"	
Bi-Temporal Change Analysis	"CDVQA (SECOND dataset test2 split) 5, 24, 40, 42, 43", "Change-VQA Accuracy (per question category & overall), Change F1-Score 40, 44"	

Optical-SAR Cross-Modal Fusion,"BigEarthNet.txt S1/S2 pairs, SOMA-1M, MM-OVSeg
CMU-Data 5, 7, 16, 29","Mean Intersection over Union (mIoU), Complementarity Tag
Agreement 15, 41"
Calibration Quality,"Held-out validation split 29, 30",Expected Calibration Error (ECE) 29

System Latency & Performance SLAs (P95 Targets)

Operation,Latency SLA (P95)

Single-Image VQA / Captioning, $\ll(< 3.0 \text{ seconds})\ll$ 46

Region Grounding, $\ll(< 4.0 \text{ seconds})\ll$ 46

Bi-Temporal Change Analysis (2 Images), $\ll(< 6.0 \text{ seconds})\ll$ 46

"Optical-SAR Fusion (2 Images, 3 Steps)", $\ll(< 8.0 \text{ seconds})\ll$ 46

"Sequential Compound Plan (e.g., Fusion $\ll(\rightarrow)\ll$ Change)",Sum of component
budgets $\ll(+ 0.5\text{s})\ll$ routing overhead 46

 *Would you like to examine the Pydantic schemas for the ExecutionTrace and
EvidenceLedger, or explore the unit-testing suite for the CompatibilityValidator?*